

Programming Techniques –Using C Language

Objective: The purpose of this lab is to become familiar with Arrays and Strings

Arrays and Strings

1. Write a program to read marks for 5 subjects using an array and compute the summation and the average of those marks.
2.
 - a. Write a program that asks the user to type 10 integers of an array.
 - b. And then read another integer variable called V, search V is in the array of 10 integers created in section (a).
 - c. If founds then program writes "V is in the array" otherwise "V is not in the array".
3. Write a program that asks the user to type 10 integers of an array. The program must output the largest element in the array, and the index at which that element was found.
4. Write a program to calculate and stores the squares of the first 10 positive integers in an array. Then display the output as follows



The square of 1 is 1
The square of 2 is 4
The square of 3 is 9
The square of 4 is 16
The square of 5 is 25
The square of 6 is 36
The square of 7 is 49
The square of 8 is 64
The square of 9 is 81
The square of 10 is 100

5. Write a program which takes 2 arrays of 10 integers each, a and b. c is an array with 20 integers. The program should put into c the appending of a and b, the first 10 integers of c from array a, the latter 10 from b. Then the program should display array c.

6. Compile and execute the following program and check the output.

```
#include <stdio.h>

int main(){
    int row, column;
    int my_array[3][3] = {{10, 23, 42},{1, 654, 0}, {652, 22, 0}};
    for (row = 0; row <=2; row++){
        for (column = 0; column <= 2; column++){
            printf("%d\t", my_array[row][column]);
            printf("\n");
        }
        printf("\n");
    }
    return 0;
}
```

7. Write a program to read your first name and display it.

Hint: define a character array to store the first name. Output should display as given below.

Enter your Name: Kamal

Hello, Kamal

8. Type the given programs and test the output.

```
#include<stdio.h>
main()
{
    char a[27];
    int i;
    for(i=65;i<91;i++)
        a[i-'A']=i;
    a[26]='\0';
    printf("%s",a);
    return 0;
}
```

```
#include <stdio.h>

main ( ){
    char word [ ] = " f i r e ";
    word [ 0 ] = 'h ' ;
    printf ( "%s \n",word );
}
```

Run it with the following substitutions added before the printf,

- word[1]='e';
- word[3]='d';

9. Write a C program to maintain a name list and check the existence of the given name in that list.
10. Write a C program to input data into two dimensional arrays and output the diagonal elements of that array.

11. Write a program that reads a user's first and last name separately. Then output the full name as a single string, for example "Nimal Perera".
- Include <string.h> header file to your program.
 - Declare two variables from suitable data type to store the first name and the last name separately.
 - Declare third variable from the same data type to store the full name. (it must be long enough to store first and last names and the separator)
 - In the full name first name and the last name must separate with the space.
 - Use strcpy(destination,source), strcat(destination, source) string functions to suitable operations.
 - Display the length of your first name by using a suitable string function.
12. Write a c program to do the followings
- Create an array to store 15 integers.
 - Store given integers to that array.
2,5,1,7,1,2,8,2,6,2,9,4,2,1,10
 - Count the number of 1 and 2's in that array.
 - Count the number of integers greater or equal to 6.
13. Write a C program to do the followings.
- Declare two arrays as follows.
 - int arr1[]={1,3,5,7,9}
 - int arr2[]={2,4,6,8,10}
 - declare another array as follows
 - int arr3[10];
 - Copy the contents of first two arrays to the third array. Finally third array should contain data as given bellow.
{1,2,3,4,5,6,7,8,9,10}
 - Display the content of third array.